People Modelling.AI — Digital Value Framework

> **Executive summary.** People Modelling.AI turns TNB GENCO's succession planning from a manual, multi-day document review into a repeatable, auditable pipeline — codifying TNB's own grade-band and discipline rules in software, and running entirely inside TNB's Databricks workspace so talent data never leaves the tenant.

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1. Objective

Level	Objective
Primary	Recommend and rank internal succession candidates for a given position, with AI-predicted readiness and written justification
Strategic	Replace spreadsheet-and-PDF talent review with a standardised, defensible process that retains TNB's succession rules as institutional software rather than individual expertise
Operational	Centralise job descriptions, talent cards, position profiles and the people model in one store; match in both directions (position to people, person to positions); log every decision for audit

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2. Function

Measured scope (excludes the superseded Streamlit prototype and all build output): **2,293 lines** of backend Python across 16 files · **1,637 lines** of frontend TypeScript/React across 5 pages · **25 REST API endpoints** across 5 routers · **128 lines** of PostgreSQL schema · 5 functional modules.

Module	What it does
Succession Recommendation	Position to ranked candidates. Auto-detects the best-matching job description by title, grade, project and specialty token scoring; filters the talent pool by grade band and education discipline; computes a succession score; batches the shortlist to the LLM for readiness classification and reasoning
Person-to-Position	The inverse direction — given an employee, returns the top-K positions they qualify for, with the same rule filtering applied
Data Management	Chunked upload and background processing of PDF and Excel sources across four categories (job descriptions, talent cards, position profiles, people model); reconciles database rows against physical storage on every read
Analytics Dashboard	Live counts of positions by project, positions by grade, and employees by grade
Audit Log	Every login, upload, deletion and recommendation run written to an indexed `audit_logs` table with actor, module, entity and outcome

The differentiating logic is the domain rules, not the model

The LLM never sees the full talent pool. It ranks a shortlist that deterministic, TNB-specific rules have already produced — which is what makes the output defensible in a talent review:

- **Grade-band adjacency.** A hand-built map of which grades are legitimate feeders for which (`M16`, `M15`, `E16&E17`, `E14&E15`, `E12&E13` and their compact variants), including cross-family tokens `CM`/`GM`/`M`/`E`. This encodes TNB's promotion structure; no external product knows it.

- **Education discipline bucketing.** Free-text qualifications are resolved into engineering families (civil, electrical, mechanical, electronic/C&I, plus the general engineering family) and the law and finance families — covering roughly 40 discipline keywords and their abbreviations.
- **Narrow versus broad JD requirements.** A job description asking for "civil engineering" is distinguished from one asking for "engineering", so a broad requirement does not silently admit every discipline and a narrow one is not over-matched.
- **TNB talent-card parsing.** Eleven named sections are extracted from the standard talent-card PDF layout (strengths, achievements, certifications, education, work experience, skills, leadership, awards, career goals, job preference, project experience), plus grade, age, join date and retirement date.
- **Project code normalisation.** NHEP, HHFS, HESS and NENGGIRI filename prefixes are resolved to canonical project names.
- **Succession scoring.** A transparent points model over KPI history, critical-role flag and years of experience — auditable, unlike a model-generated score.

Architecture

React 19 + Vite + Tailwind single-page frontend, served by a FastAPI backend that also exposes the API. Storage is SQLite by default and PostgreSQL when configured, behind a compatibility layer, with vector-ready tables (pgvector, plus a fallback schema) already provisioned for future semantic retrieval. Inference runs against a Databricks serving endpoint (`databricks-llama-4-maverick`) using workspace identity — no third-party API keys — with a three-tier call fallback and exponential-backoff retry. Deployment targets Databricks Apps with Unity Catalog Volume storage.

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3. Effort Taken (Development Mandays)

> **Estimate to validate.** The workstream figures below are reconstructed from measured code volume, module count and the scope of domain-rule elicitation. They are not timesheet data. Confirm against the actual project record before external use.

Workstream	Mandays
Discovery and HR rule elicitation (grade bands, discipline families, talent-card layout)	12
Document parsing and ingestion pipeline (PDF/Excel, chunked upload, background processing)	22
Matching and scoring engine (education buckets, grade adjacency, JD auto-detection, succession score)	25
LLM integration (Databricks endpoint, fallback chain, prompt design, response parsing)	12
Backend API (25 endpoints, storage reconciliation, audit logging)	20
React frontend (5 pages, Excel/PDF export)	25
Database and persistence (SQLite/PostgreSQL layer, schema, vector-ready tables)	10
Databricks Apps deployment and UC Volume integration	8
Migration from Streamlit prototype (parity verification across 8 pages)	10
Testing, UAT and documentation	14
Total	**158**

At a team of 2–3, this represents roughly **3–4 months** of elapsed delivery. At an assumed blended day rate of **RM 800–1,500** — *an assumption to confirm with procurement, not a quoted rate* — the equivalent external build cost is **RM 126,000–237,000**.

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4. Market Solutions & Subscription Cost

There is no direct off-the-shelf equivalent. Every category below solves part of the problem, but none encodes TNB's grade bands, discipline rules or talent-card format — that configuration work would be required on top of any licence fee. That absence is itself a value point.

Category (example products)	Indicative cost	Gap vs People Modelling.AI
Enterprise HCM succession modules (SAP SuccessFactors, Oracle HCM, Workday)	RM 200–600 per employee per year	Generic 9-box and competency models; TNB grade-band and discipline logic must be custom-configured; vendor-cloud hosted, so talent data leaves the tenant
Standalone talent review and succession tools (Cornerstone, PeopleFluent)	RM 120–350 per employee per year	Expect clean structured HRIS records; cannot ingest the existing TNB talent-card and JD PDFs without a separate data programme
AI candidate-matching platforms (Eightfold-class, HiredScore-class)	RM 250–700 per employee per year	Oriented to external recruitment and CV matching; no internal grade-progression rules and no readiness horizon for existing staff
General-purpose LLM subscriptions (per-seat chat assistants)	RM 90–150 per seat per month	No persistence, no deterministic shortlist, no audit trail; every review is an unrepeatable one-off conversation

All figures are indicative list-price ranges for scoping only — verify in procurement before use in a business case.

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5. Productivity Improvement vs. manual process

> **Estimate to validate.** The "before" column reflects the manual process as described; the improvement percentages follow from it. Confirm the baseline with the HR team before external use.

Task	Before → After	Est. improvement
Shortlist candidates for one position	Manual read of the talent pool with grade and education cross-checked by hand, 2–3 days → rules engine returns a ranked, filtered shortlist in seconds	90–95%
Extract structured fields from a talent card	Manual transcription into a spreadsheet, 15–20 minutes per card → parsed automatically on upload	95%+
Match one person to open positions	Compare the individual against each JD in turn, around half a day → top-K positions returned in under a minute	85–95%
Draft readiness justification	Written from scratch per candidate, 30–60 minutes → AI-drafted reasoning reviewed and edited by HR	60–75%
Workforce analytics by project and grade	Rebuild pivot tables in Excel per request, 2–4 hours → dashboard reads live from the database	80–90%
Evidence trail for a talent decision	Reconstructed from email and file versions after the fact → every run logged automatically with actor, inputs and outcome	Qualitative — new capability

6. Other Value Levers

- **Knowledge retention.** TNB's grade-band adjacency and discipline rules now live in version-controlled code rather than in the heads of a few experienced HR practitioners. They survive transfers and retirements, and any change to them is reviewable.
- **Data sovereignty and security.** The application runs inside TNB's own Databricks workspace, against a workspace-hosted model serving endpoint authenticated by workspace identity. No employee talent data, KPI history or personal record is sent to an external vendor cloud or third-party API — a materially different risk posture from any SaaS HCM alternative.
- **Risk reduction.** Deterministic pre-filtering means a candidate is never recommended outside their eligible grade band or required discipline, regardless of model behaviour. AI output is confined to ranking and explanation.
- **Compliance and auditability.** The `audit_logs` table gives a defensible answer to "why was this person shortlisted and that person not", which matters for both internal governance and any external challenge.
- **Standardisation.** One shortlisting method applied identically across projects and reviewers, replacing per-reviewer judgement about what counts as a relevant qualification.
- **Vendor independence.** No per-employee licence, no seat count, no renewal negotiation. Scaling to more positions or more staff costs nothing additional.
- **Scalability.** Vector-ready schema and chunked ingestion are already in place, so semantic search over the document corpus is an increment rather than a rebuild.
- **Talent uplift.** The build leaves TNB with in-house capability in FastAPI, React and Databricks model serving, applicable to the next internal AI use case.

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7. Readiness & Next Steps

Stated plainly, so the framework is not mistaken for a production sign-off:

- **Authentication is a placeholder.** A single hardcoded credential pair returns a static token, and no API endpoint validates it; CORS is currently open. Integrating Databricks Apps identity or TNB SSO, and enforcing it per endpoint, is the prerequisite for handling live talent data.
- **Role-based access is not yet implemented.** Every authenticated user has full rights, including deletion. HR reviewer, project owner and administrator roles should be separated before wider rollout.
- **The vector tables are provisioned but unused.** Semantic retrieval over job descriptions and talent cards is the natural next increment.
- **Validate the estimates in sections 3–5** against project records and the HR baseline before this document is used in a business case.

Headline for stakeholders: *An internally built succession engine that encodes TNB's own promotion rules in software, cuts position shortlisting from days to seconds, and keeps every talent record inside TNB's tenant — at a one-off build cost in the region of RM 126,000–237,000 rather than a perpetual per-employee licence.*